# Day 33

## The Error Object & Custom Errors:

**What is the Error Object?**

In JavaScript, whenever something goes wrong, the language creates an Error object. This object contains useful information about what went wrong.

The Error object has important properties:

| Property | Meaning |
|---|---|
| **name** | Type of error (Error, ReferenceError, TypeError, etc.) |
| **message** | Description of the problem |
| **stack** | Where the error happened (line number) |

**Built-in Error Types (common)**

| Error Type | When it happens |
|---|---|
| **ReferenceError** | Using a variable that doesn't exist |
| **TypeError** | Using a wrong data type (calling a number like a function) |
| **SyntaxError** | Code written incorrectly (usually caught before running) |
| **RangeError** | Number out of allowed range (stack overflow) |
| **Error** | Generic error |

Example: a basic example showing try...catch.

```
JS main.js > ...
try {
    Aditya;
} catch (error) {
    console.log(error);
}
```

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```
PS E:\JavaScript\Day31-40\Day33> node main.js
ReferenceError: Aditya is not defined
    at Object.<anonymous> (E:\JavaScript\Day31-40\Day33\main.js:2:5)
```

Now, getting the name and message for an error:

```
JS main.js > ...
try {
    Aditya;
} catch (error) {
    console.log(error.name);
    console.log(error.message);
}
```

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```
PS E:\JavaScript\Day31-40\Day33> node main.js
ReferenceError
Aditya is not defined
```

**Creating Custom Errors**

You can create your own errors using the Error constructor — useful for validation, API responses, and logic checks.

Example:

```
//Custom error
function login(user) {
    if (!user) {
        throw new Error("User not found!");
    }
    return "Logged in";
}

try {
    login(null);
} catch (err) {
    console.log(err.message); // User not found!
}
```

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```
PS E:\JavaScript\Day31-40\Day33> node main.js
User not found!
```

Example: throwing errors manually

```
function divide(a, b) {
    if (b === 0) {
        throw new Error("Cannot divide by zero");
    }
    return a / b;
}
```

--The End--